

1. Array Store & Print

- Write a program in C to store elements in an array and print them.

Test Data :

Input 10 elements in the array :

element - 0 : 1

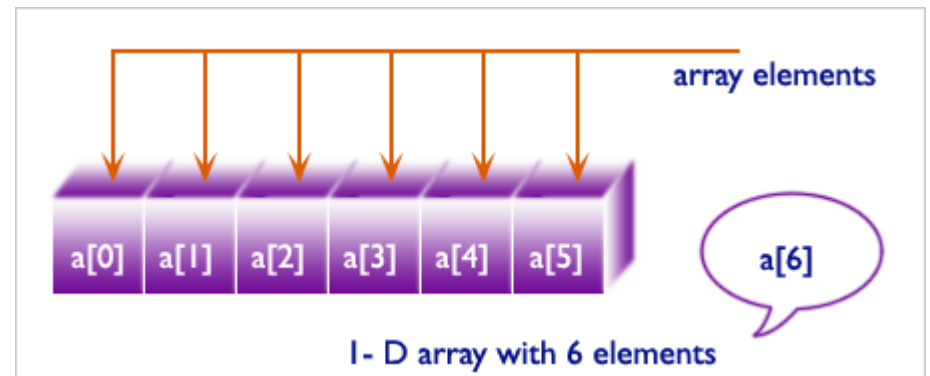
element - 1 : 1

element - 2 : 2

.....

- *Expected Output :*

Elements in array are: 1 1 2 3 4 5 6 7 8 9



```
1  #include <stdio.h>
2  // Main function
3  int main()
4  {
5      int arr[10]; // Declare an array of size 10 to store integer values
6      int i;
7      // Print a message to prompt the user for input
8      printf("\n\nRead and Print elements of an array:\n");
9      printf("-----\n");
10     // Prompt the user to input 10 elements into the array
11     printf("Input 10 elements in the array :\n");
12     for(i=0; i<10; i++)
13     {
14         printf("element - %d : ",i); // Prompt the user to input the i-th element
15         scanf("%d", &arr[i]); // Read the input and store it in the array
16     }
17     // Display the elements in the array
18     printf("\nElements in array are: ");
19     for(i=0; i<10; i++)
20     {
21         printf("%d ", arr[i]); // Print each element in the array
22     }
23     printf("\n");
24     return 0;
25 }
```

2. Array Reverse Display

- Write a program in C to read n number of values in an array and display them in reverse order.

Test Data :

Input the number of elements to store in the array :3

Input 3 number of elements in the array :

element - 0 : 2

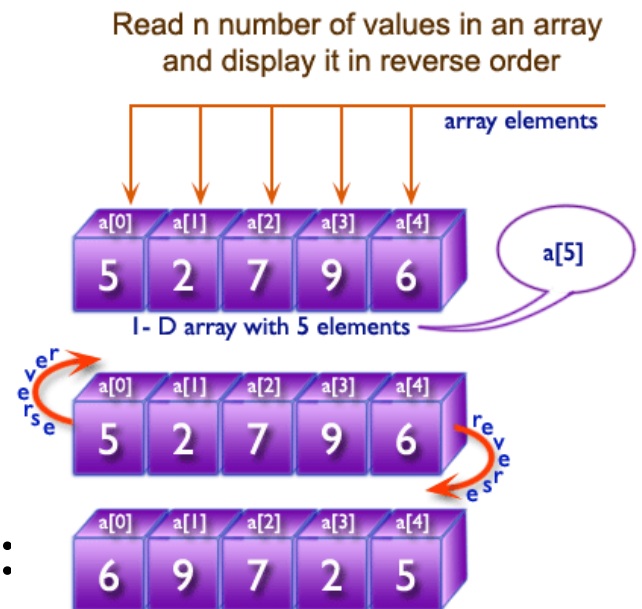
element - 1 : 5

element - 2 : 7

Expected Output :

The values store into the array are :
2 5 7

The values store into the array in reverse are :
7 5 2



```
1  #include <stdio.h>
2  // Main function
3  int main()
4  {
5      int i, n, a[100];
6      // Display a message to the user about the program's purpose
7      printf("\n\nRead n number of values in an array and display it in reverse order:\n");
8      printf("-----\n");
9      // Prompt the user to input the number of elements to store in the array
10     printf("Input the number of elements to store in the array :");
11     scanf("%d", &n);
12     // Prompt the user to input n elements into the array
13     printf("Input %d number of elements in the array :\n", n);
14     for (i = 0; i < n; i++)
15     {
16         printf("element - %d : ", i);
17         scanf("%d", &a[i]); // Read the input and store it in the array
18     }
19     // Display the values stored in the array
20     printf("\nThe values stored in the array are : \n");
21     for (i = 0; i < n; i++)
22     {
23         printf("% 5d", a[i]); // Print each element in the array
24     }
25     // Display the values stored in the array in reverse order
26     printf("\n\nThe values stored in the array in reverse are :\n");
27     for (i = n - 1; i >= 0; i--)
28     {
29         printf("% 5d", a[i]); // Print each element in reverse order
30     }
31     printf("\n\n");
32     return 0;
33 }
```

3. Array Copy

- Write a program in C to copy the elements of one array into another array.

Test Data :

Input the number of elements to be stored in the array :3

Input 3 elements in the array :

element - 0 : 15

element - 1 : 10

element - 2 : 12

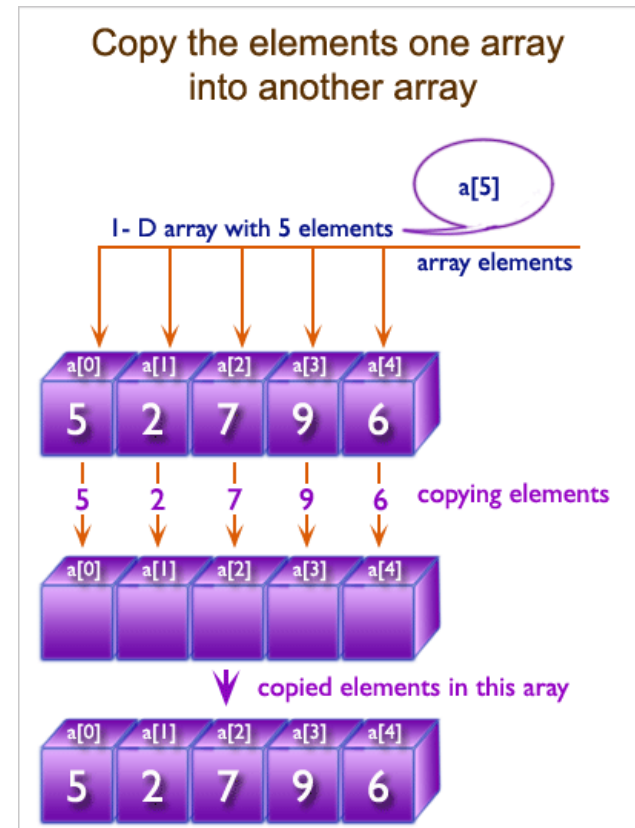
Expected Output :

The elements stored in the first array are :

15 10 12

The elements copied into the second array are :

15 10 12



```
1  #include <stdio.h>
2  // Main function
3  int main()
4  {
5      int arr1[100], arr2[100]; // Declare two arrays of size 100 to store integer values
6      int i, n; // Declare variables to store array size and loop counter
7      // Display a message to the user about the program's purpose
8      printf("\n\nCopy the elements of one array into another array:\n");
9      printf("-----\n");
10     // Prompt the user to input the number of elements to be stored in the array
11     printf("Input the number of elements to be stored in the array :");
12     scanf("%d", &n);
13     // Prompt the user to input n elements into the first array
14     printf("Input %d elements in the array :\n", n);
15     for (i = 0; i < n; i++)
16     {
17         printf("element - %d : ", i);
18         scanf("%d", &arr1[i]); // Read the input and store it in the first array
19     }
20     /* Copy elements of the first array into the second array. */
21     for (i = 0; i < n; i++)
22     {
23         arr2[i] = arr1[i]; // Copy each element from the first array to the second array
24     }
25 }
```

```
26      /* Prints the elements of the first array */
27      printf("\nThe elements stored in the first array are :\n");
28      for (i = 0; i < n; i++)
29      {
30          printf("% 5d", arr1[i]); // Print each element in the first array
31      }
32      /* Prints the elements copied into the second array. */
33      printf("\n\nThe elements copied into the second array are :\n");
34      for (i = 0; i < n; i++)
35      {
36          printf("% 5d", arr2[i]); // Print each element in the second array
37      }
38      printf("\n\n"); // Print new lines for better formatting
39      return 0;
40  }
```

4. Merge Two Sorted Arrays (Descending)

- Write a program in C to merge two arrays of the same size sorted in descending order.

Test Data :

Input the number of elements to be stored in the first array :3

Input 3 elements in the array :

element - 0 : 1

element - 1 : 2

element - 2 : 3

Input the number of elements to be stored in the second array :3

Input 3 elements in the array :

element - 0 : 1

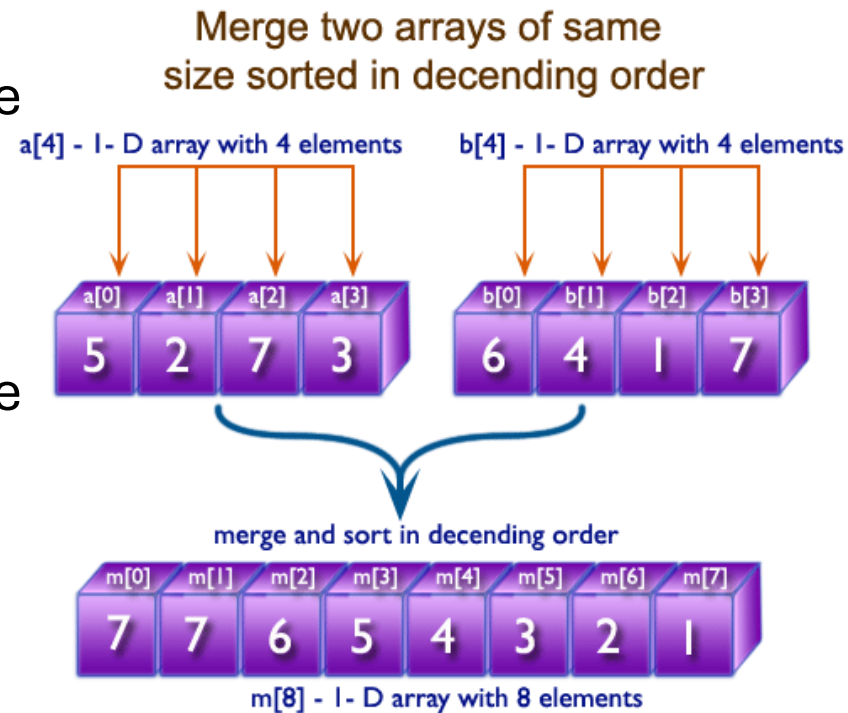
element - 1 : 2

element - 2 : 3

Expected Output :

The merged array in descending order is :

3 3 2 2 1 1




```
1  #include <stdio.h>
2
3  int main()
4  {
5      int arr1[100], arr2[100], arr3[200];
6      int s1, s2, s3;
7      int i, j, k;
8
9
10     printf("\n\nMerge two arrays of same size sorted in decending order.\n");
11     printf("-----\n");
12
13     printf("Input the number of elements to be stored in the first array :");
14     scanf("%d",&s1);
15
16     printf("Input %d elements in the array :\n",s1);
17     for(i=0;i<s1;i++)
18     {
19         printf("element - %d : ",i);
20         scanf("%d",&arr1[i]);
21     }
22
```

```
23
24     printf("Input the number of elements to be stored in the second array :");
25     scanf("%d",&s2);
26
27     printf("Input %d elements in the array :\n",s2);
28     for(i=0;i<s2;i++)
29     {
30         printf("element - %d : ",i);
31         scanf("%d",&arr2[i]);
32     }
33
34     /* size of merged array is size of first array and size of second array */
35     s3 = s1 + s2;
36     /*----- insert in the third array-----*/
37     for(i=0;i<s1; i++)
38     {
39         arr3[i] = arr1[i];
40     }
41     for(j=0;j<s2; j++)
42     {
43         arr3[i] = arr2[j];
44         i++;
45     }
```

```
46  /*----- sort the array in decending order -----*/
47  for(i=0;i<s3; i++)
48      {
49          for(k=0;k<s3-1;k++)
50              {
51
52                  if(arr3[k]<=arr3[k+1])
53                      {
54                          j=arr3[k+1];
55                          arr3[k+1]=arr3[k];
56                          arr3[k]=j;
57                      }
58              }
59      }
60
61  /*----- Prints the merged array -----*/
62  printf("\nThe merged array in decending order is :\n");
63  for(i=0; i<s3; i++)
64      {
65          printf("%d  ", arr3[i]);
66      }
67  printf("\n\n");
68  return 0;
69 }
```

5. Frequency of Array Elements

- Write a program in C to count the frequency of each element of an array.

Test Data :

Input the number of elements to be stored in the array :3

Input 3 elements in the array :

element - 0 : 25

element - 1 : 12

element - 2 : 43

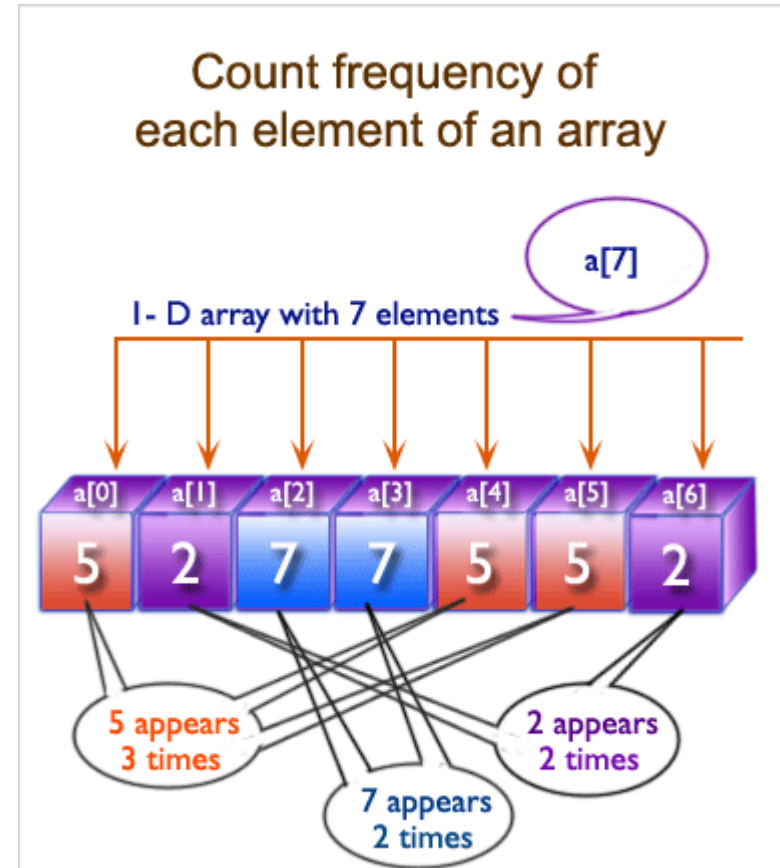
Expected Output :

The frequency of all elements of an array :

25 occurs 1 times

12 occurs 1 times

43 occurs 1 times



```
1  #include <stdio.h>
2
3  int main()
4  {
5      int arr1[100], fr1[100];
6      int n, i, j, ctr;
7
8      // Prompt user for input
9      printf("\n\nCount frequency of each element of an array:\n");
10     printf("-----\n");
11     printf("Input the number of elements to be stored in the array :");
12     scanf("%d", &n);
13
14     // Input elements for the array
15     printf("Input %d elements in the array :\n", n);
16     for (i = 0; i < n; i++)
17     {
18         printf("element - %d : ", i);
19         scanf("%d", &arr1[i]);
20         fr1[i] = -1; // Initialize frequency array with -1
21     }
22
```

```
23 // Count the frequency of each element in the array
24 for (i = 0; i < n; i++)
25 {
26     ctr = 1; // Initialize counter for each element
27     for (j = i + 1; j < n; j++)
28     {
29         if (arr1[i] == arr1[j])
30         {
31             ctr++; // Increment counter for matching elements
32             fr1[j] = 0; // Mark the duplicate element's frequency as 0
33         }
34     }
35
36     // If frequency array value is not marked as 0, set it to the counter
37     if (fr1[i] != 0)
38     {
39         fr1[i] = ctr;
40     }
41 }
42
```

```
43 // Print the frequency of each element in the array
44 printf("\nThe frequency of all elements of the array : \n");
45 for (i = 0; i < n; i++)
46 {
47     if (fr1[i] != 0)
48     {
49         printf("%d occurs %d times\n", arr1[i], fr1[i]);
50     }
51 }
52 return 0;
53 }
```

6. Separate Odd & Even Arrays

- Write a program in C to separate odd and even integers into separate arrays.

Test Data :

Input the number of elements to be stored in the array :5

Input 5 elements in the array :

element - 0 : 25

element - 1 : 47

element - 2 : 42

element - 3 : 56

element - 4 : 32

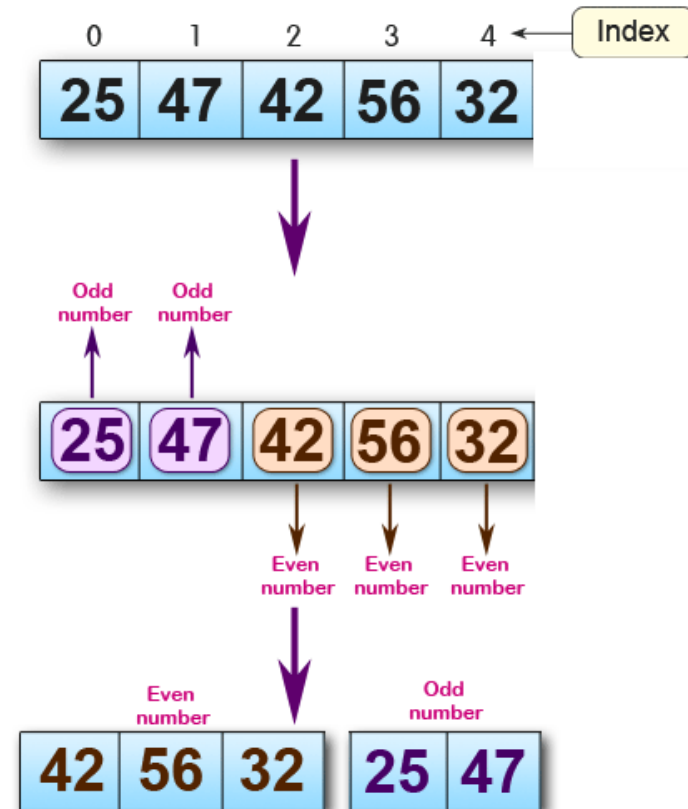
Expected Output :

The Even elements are :

42 56 32

The Odd elements are :

25 47



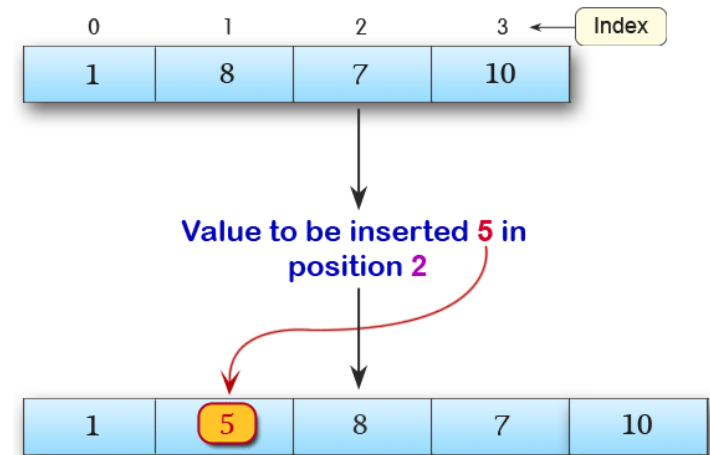

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int arr1[10], arr2[10], arr3[10];
6      int i, j = 0, k = 0, n;
7
8      // Prompt user for input
9      printf("\n\nSeparate odd and even integers in separate arrays:\n");
10     printf("-----\n");
11     printf("Input the number of elements to be stored in the array :");
12     scanf("%d", &n);
13
14     // Input elements for the array
15     printf("Input %d elements in the array :\n", n);
16     for (i = 0; i < n; i++)
17     {
18         printf("element - %d : ", i);
19         scanf("%d", &arr1[i]);
20     }
21
```

```
22     // Separate odd and even integers into separate arrays
23     for (i = 0; i < n; i++)
24     {
25         if (arr1[i] % 2 == 0)
26         {
27             arr2[j] = arr1[i];
28             j++;
29         }
30         else
31         {
32             arr3[k] = arr1[i];
33             k++;
34         }
35     }
36
```

```
37     // Print the Even elements
38     printf("\nThe Even elements are : \n");
39     for (i = 0; i < j; i++)
40     {
41         printf("%d ", arr2[i]);
42     }
43
44     // Print the Odd elements
45     printf("\nThe Odd elements are :\n");
46     for (i = 0; i < k; i++)
47     {
48         printf("%d ", arr3[i]);
49     }
50     printf("\n\n");
51     return 0;
52 }
```

7. Insert in Unsorted Array

- Write a program in C to insert values in the array (unsorted list).
Test Data :
Input the size of array : 4
Input 4 elements in the array in ascending order:
element - 0 : 1
element - 1 : 8
element - 2 : 7
element - 3 : 10
Input the value to be inserted : 5
Input the Position, where the value to be inserted : 2
Expected Output :
The current list of the array :
1 8 7 10
After Insert the element the new list is :
1 5 8 7 10



```
1  #include <stdio.h>
2
3  void main()
4  {
5      int arr1[100], i, n, p, x;
6
7      // Prompt user for input
8      printf("\n\nInsert New value in the unsorted array:\n ");
9      printf("-----\n");
10     printf("Input the size of array : ");
11     scanf("%d", &n);
12
13     // Input values for the array
14     printf("Input %d elements in the array in ascending order:\n", n);
15     for (i = 0; i < n; i++)
16     {
17         printf("element - %d : ", i);
18         scanf("%d", &arr1[i]);
19     }
20
21     // Input the value to be inserted and its position
22     printf("Input the value to be inserted : ");
23     scanf("%d", &x);
24     printf("Input the Position, where the value to be inserted :");
25     scanf("%d", &p);
26
```

```
27 // Display the current list of the array
28 printf("The current list of the array :\n");
29 for (i = 0; i < n; i++)
30     printf("% 5d", arr1[i]);
31
32 // Move all data to the right side of the array to make space
33 for (i = n; i >= p; i--)
34     arr1[i] = arr1[i - 1];
35
36 // Insert the new value at the given position
37 arr1[p - 1] = x;
38
39 // Display the array after insertion
40 printf("\n\nAfter Insert the element the new list is :\n");
41 for (i = 0; i <= n; i++)
42     printf("% 5d", arr1[i]);
43 printf("\n\n");
44 }
```

8. Delete Element from Array

- Write a program in C to delete an element at a desired position from an array.

Test Data :

Input the size of array : 5

Input 5 elements in the array in ascending order:

element - 0 : 1

element - 1 : 2

element - 2 : 3

element - 3 : 4

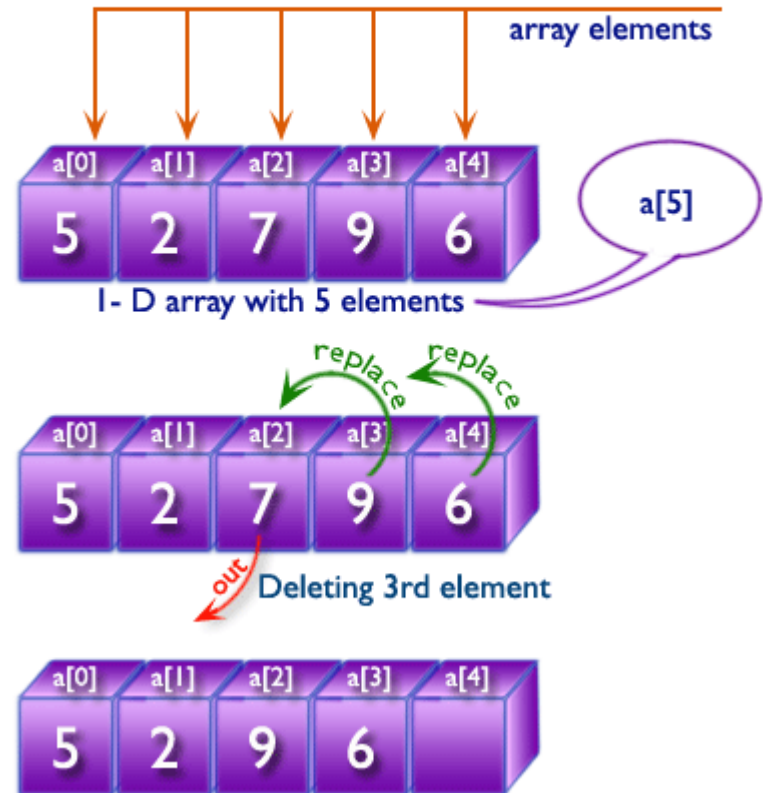
element - 4 : 5

Input the position where to delete: 3

Expected Output :

The new list is : 1 2 4 5

Delete an element at desired position from an array



```
1  #include <stdio.h>
2
3  void main() {
4      int arr1[50], i, pos, n;
5
6      // Prompt user for input
7      printf("\n\nDelete an element at the desired position from an array:\n");
8      printf("-----\n");
9      printf("Input the size of the array : ");
10     scanf("%d", &n);
11
12     // Input values for the array
13     printf("Input %d elements in the array in ascending order:\n", n);
14     for (i = 0; i < n; i++) {
15         printf("element - %d : ", i);
16         scanf("%d", &arr1[i]);
17     }
18
19     // Input the position to delete
20     printf("\nInput the position where to delete: ");
21     scanf("%d", &pos);
22 }
```



```
23 // Locate the position in the array
24 i = 0;
25 while (i != pos - 1)
26     i++;
27
28 // Shift elements to the left to replace the deleted element
29 while (i < n) {
30     arr1[i] = arr1[i + 1];
31     i++;
32 }
33 n--;
34
35 // Display the new list after deletion
36 printf("\nThe new list is : ");
37 for (i = 0; i < n; i++) {
38     printf(" %d", arr1[i]);
39 }
40 printf("\n\n");
41 }
```